When “Try Harder” Backfires: Agent-Based Simulation of Tone-Dependent Attrition in Diet Coaching

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***Abstract*— Commercial nutrition apps face persistent adherence challenges: users often disengage after routine disruptions, and failure-style corrective feedback can amplify shame and dropout for vulnerable users. Because evaluating punitive interaction policies in human studies is ethically constrained and slow, we use simulation-based testing of coaching strategies before deployment. We present CARE-Sim (Compassionate AI Response Evaluation Simulator), a multi-agent framework that instantiates LLM-driven digital twin users with memory and evolving internal state to compare rigid versus compassionate diet-coaching policies over a fixed horizon. We simulated N = 400 agents stratified into four vulnerability-informed cohorts (100 per cohort) and ran R = 5 independent 60-day simulations per policy, yielding 2,000 traces per policy. Attrition is not imposed as a predefined curve; instead, dropout emerges when an agent’s internal motivation state crosses a threshold updated by daily affective shifts inferred from reflective self-talk. Across cohorts, compassionate coaching increased Day-60 retention relative to rigid coaching, with the largest advantages for Perfectionist (+31.3 percentage points) and Habitual/Defensive (+28.0 percentage points) cohorts. Mechanistically, compassionate coaching weakened the coupling between dietary error and negative self-talk (rigid: r = 0.82; compassionate: r = 0.35) and was associated with fewer traces consistent with counter-regulatory responding. We position CARE-Sim as a hypothesis-generating tool for human-factors evaluation of coaching tone, enabling safer pre-deployment comparison of interaction policies and targeted, user-centered validation.**

Keywords—*Large Language Models, Digital Twins, Behavioral Simulation, mHealth, Adherence.*

I. Introduction

Commercial nutrition and diet interventions continue to face a persistent adherence problem: the physiological logic is straightforward, but day-to-day follow through is fragile. Users often begin with high intent, then disengage after routine disruptions (e.g., travel, workload, stress, illness) that break logging habits and goal pursuit. Attrition in digital health is common enough to be framed as a “law of attrition,” and adherence has been shown to depend on specific design choices rather than intent alone [1], [2]. Many systems, however, still treat deviations as binary failure states and return generic corrective feedback, despite well-established differences in how individuals appraise setbacks and regulate negative affect [3]. These differences can have consequential downstream effects.

Rigid feedback can be disproportionately harmful for users who are high in perfectionism, self-criticism, or negative affectivity. A short warning-style message (“noncompliant,” “failed,” “try harder”) may function as a motivator for one user but as a moralized judgment for another, converting a single lapse into a global self-assessment (“I can’t do this”), consistent with shame-maintenance dynamics reported in behavior-change and addiction contexts [4]. A 'shame spiral' occurs when a single behavioral lapse triggers intense self-criticism, leading to a loss of self-efficacy and subsequent, more severe lapses. 'Psychological reactance' refers to an individual's oppositional behavioral response—such as intentionally overeating—when they perceive a rigid warning as an unwarranted threat to their autonomy. Once that boundary is perceived as crossed, counter-regulatory behavior can follow—the “what-the-hell” pattern in which a deviation triggers additional deviation rather than recovery [5]. In human-factors terms, the issue is not whether a lapse is detected accurately, but whether the interaction policy fits the user’s vulnerability profile.

Large language models (LLMs) expand the design space for adaptive feedback by generating context-sensitive language that users often perceive as more empathetic and supportive. In a controlled comparison using patient questions, chatbot responses were rated higher than physician responses in perceived empathy and overall quality [6], suggesting that “engineered empathy” can be treated as a system capability rather than a fixed attribute. Yet evaluating punitive or shame-inducing interaction policies in human studies is ethically constrained, and iterative tone experimentation is slow. We

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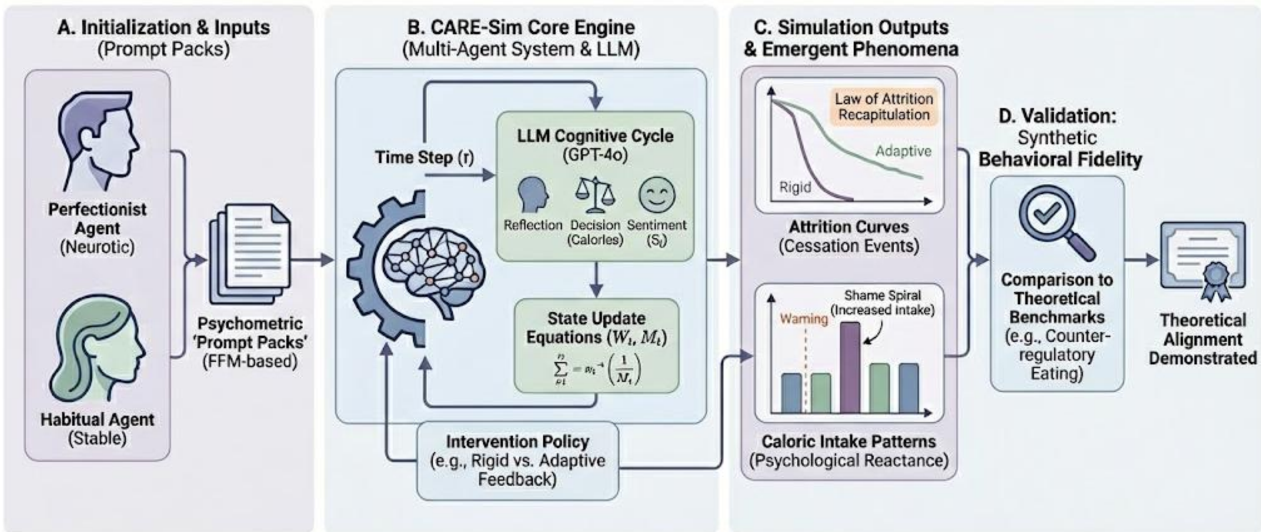

Fig. 1. An Overview of the CARE-Sim Methodology and Validation Framework. It consists of modules, including psychometric agent initialization via prompt packs, an LLM-driven simulation engine, and state updates, and the generation of emergent behavioral traces for validation against established psychological theories.

therefore introduce CARE-Sim (Compassionate AI Response Evaluation Simulator), a multi-agent framework that instantiates LLM-driven "digital twin" users with memory and evolving internal state. CARE-Sim compares rigid versus compassionate feedback policies over a fixed horizon. Digital twins have been proposed in health to simulate trajectories and probe intervention strategies prior to deployment [7][8]. CARE-Sim adapts this paradigm for human factors evaluation by allowing dropout and behavioral drift to emerge from state dynamics rather than being imposed as a predetermined attrition curve.

The primary aim of this study is to explore whether agent-based simulation can serve as a safe, in silico testbed for evaluating the behavioral impacts of varying diet-coaching feedback tones. Specifically, we ask: How does a compassionate versus a rigid feedback policy influence user attrition across different psychometric profiles in a 60-day simulated environment? We position CARE-Sim not as a substitute for human trials, but as a pre-clinical screening tool—similar to computational models used in drug discovery. To be considered trustworthy, CARE-Sim must demonstrate qualitative alignment with established behavioral theories. Its utility lies in flagging potentially harmful interaction policies before deployment, thereby narrowing the design space and mitigating ethical risks. Any policy deemed beneficial in simulation must subsequently undergo targeted, user-centered human validation.

II. Methodology

A. Digital Twin Architecture

We implemented CARE-Sim as a multi-agent simulation in which each "digital twin" is a persistent computational user model with memory and an evolving internal state (**Fig. 1**). In this study, we instantiated N = 400 unique agents (100 per psychometric cohort). For each feedback policy condition (**Rigid** vs. **Compassionate**), we performed 5 independent 60-day simulation runs, each with a different random seed used to control stressor schedules and stochastic sampling.

- **Agent**: a unique parameterized user profile (fixed across runs).
- **Run**: one 60-day simulated episode for all 400 agents under one policy with a specific seed.
- **Trace**: one agent's 60-day record within a run.

Thus, each policy condition produced 400 agents × 5 runs = 2,000 traces, and the full study produced 4,000 traces across both policies (unless otherwise stated). Each simulated day produces a structured record (the CARE-Sim Run Log) containing: (i) the stressor event, (ii) the agent's caloric decision, (iii) the app feedback message, (iv) the agent's reflective monologue, (v) sentiment score, and (vi) whether a cessation (dropout) event occurred.

We selected a 60-day simulation horizon to capture both initial engagement and the long-tail dropout characteristic of digital health apps. A 7-day memory window was chosen to reflect human working memory constraints regarding recent dietary habits, avoiding the context-window hallucination risks of unbounded LLM memory while capturing weekly behavioral cycles. To capture human variability and address simplistic attrition rules, stochastic noise (ϵ) is incorporated into the daily caloric decision

Agents were orchestrated in LangChain and equipped with a "MemoryGraph" that maintains a longitudinal history of adherence-relevant events and prior app interactions. The memory store supports two retrieval modes: (i) recent-window recall (last **k = 7** days of logs) and (ii) salient-episode recall (e.g., prior high-deviation or high-conflict days) using a rule-based salience filter keyed to adherence-relevant tags in the log.

The language model used for agent reasoning and generation was **OpenAI GPT-4o** (model identifier recorded per run). Each

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run records sampling settings (e.g., temperature), the prompt versions used ("Prompt Pack"), and the per-day random seed.

B. Psychometric Cohort Stratification

Neuroticism and self-compassion were selected together because they represent primary drivers of emotional reactivity and recovery in dietary contexts. Neuroticism dictates the initial stress response to a lapse, while self-compassion governs the cognitive reframing necessary to prevent a lapse from becoming a full relapse. We derived four distinct cohorts to represent a spectrum of user vulnerabilities: 'Perfectionist' (high stress sensitivity, low recovery), 'Habitual/Defensive' (moderate stress sensitivity, high negative appraisal), 'Resilient' (low stress sensitivity, high recovery), and 'Control' (baseline values). The specific parameter values (M_0, λ, α, , β) were heuristically chosen to encode these theoretical extremes, establishing a diverse synthetic population for stress-testing. While the current model decouples reflection status (S) and attrition (M) from baseline trait motivation to strictly isolate the impact of daily feedback tone on sentiment, future iterations will explicitly moderate cognitive reflection depths using these baseline individual attributes.

To stress-test feedback policies against user variability, we stratified agents into four profiles using constructs related to neuroticism (*Five-Factor Model*) and self-compassion (*Self-Compassion Scale*). Each cohort is operationalized via four coefficients written into the agent profile at initialization:

- **Baseline motivation**: $\boldsymbol{M_0}$
- **Stress sensitivity** ($\boldsymbol{\lambda}$): how strongly stress degrades willpower
- **Negative appraisal gain** ($\boldsymbol{\alpha}$): how strongly rigid feedback amplifies negative self-talk
- **Recovery gain** ($\boldsymbol{\beta}$): how strongly compassionate feedback supports reframing/recovery

Cohort parameters used in this study were:

- **Perfectionist**: M_0 = 85, λ = 0.9, $\alpha = 1.5, \beta = 0.2$
- **Habitual/Defensive**: M_0 = 60, λ = 0.8, $\alpha = 1.2$, β=0.4
- **Resilient**: M_0 = 75, $\lambda = 0.5, \alpha = 0.5, \beta = 0.9$
- **Control**: M_0 = 70, λ = 0.5, α = 1.0, β = 0.5

C. Simulation Loop

Each run simulates **60 consecutive days** of app use for each agent under a fixed feedback policy (Rigid or Compassionate). Each day t consists of four steps:

Step 1. Stressor generation (World Agent).

A "World Agent" introduces a barrier event sampled from a fixed taxonomy of stressors. Each stressor is represented as a tuple where intensity lies on a bounded scale (e.g., [0,1]). The run seed determines the daily stressor schedule.

Step 2. Caloric decision (User Agent).

Given the stressor and the user agent's current internal state, the user agent produces a caloric intake decision for day t. We maintain a bounded willpower state W_t in [0,1] updated via:

$$W_t = clip(W_{t-1} - \lambda \cdot Stress_t + \rho \cdot Recovery_{t-1},\ 0,1) \quad (1)$$

where λ is cohort-specific stress sensitivity, ρ is a recovery coefficient, and $Recovery_{t-1}$ is derived from the prior day's compassionate reframing signal under the Compassionate policy (and is 0 under the Rigid policy).

$$Calories_t = \text{Target} + \epsilon_t - \gamma\, W_t \qquad (2)$$

where ϵ_t is stochastic noise (e.g., $\epsilon_t \sim \mathrm{N}\,(0,\sigma^2)$) and γ controls the degree to which willpower suppresses overeating.

'Target' refers to the daily prescribed caloric intake limit. We simulated 400 agents (100 per cohort) over 5 independent runs to balance statistical stability with computational cost. Preliminary variance checks indicated that 5 runs were sufficient to achieve high run-to-run consistency (Cronbach's $\alpha = 0.88$) in cohort-level retention trajectories.

Step 3. App feedback (App Agent).

The "App Agent" delivers feedback using one of two policy prompts. In the Rigid condition, feedback is framed as a warning tied to numeric exceedance (e.g., "Limit exceeded by 300 kcal"). In the Compassionate condition, feedback uses reframing and normalizes lapse under stress (e.g., "A high-stress day—let's get back on track tomorrow."). Policy prompts are versioned and recorded in the Prompt Pack.

TABLE I. SUMMARY OF USER RETENTION OUTCOMES AT DAY 60 BY PERSONALITY COHORT AND COACHING STYLE.

Cohorts	**User Retention Outcomes**			
	Rigid (Day 60)	***Compassionate (Day 60)***	***Difference***	***Divergence Onset***
Habitual	8.7% [5.3–12.0]	36.7% [32.0–41.3]	+28.0 pp	Day 4
Perfectionist	24.7% [18.6–30.7]	56.0% [53.6–58.4]	+31.3 pp	Day 9
Control	46.7% [40.5–52.9]	61.3% [56.9–65.8]	+14.7 pp	Day 16
Resilient	76.7% [73.1–80.2]	89.3% [84.5–94.1]	+12.7 pp	Day 52

Step 4. Reflection and sentiment scoring.

After receiving feedback, the user agent generates a brief internal monologue capturing self-talk and intentions for tomorrow. We compute a valence score S_t using VADER sentiment analysis on the monologue text. We use the VADER

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compound score (range [−1,1]) as S_t, and define the daily sentiment shift:

$$\Delta S_t = S_t - S_{\{t-1\}} \tag{3}$$

D. Emergent Attrition (Cessation) Logic

Attrition is not imposed as a fixed curve. Instead, agents exit when an internal motivation state crosses a threshold. We maintain a scalar motivation score M_t updated by daily sentiment shift:

$$M_t = M_{\{t-1\}} + \Delta S_t \tag{4}$$

A cessation event (dropout) occurs on the first day when $M_t \leq 0$. The Run Log records (M_t, S_t, ΔS_t) per day, enabling sensitivity analyses on the sentiment model and cessation threshold.

E. Summary Metrics and Confidence Intervals

We report retention and behavioral summaries at the cohort × policy level. To avoid overstating inference from synthetic data, we treat uncertainty primarily as run-to-run stability across the R=5 independent runs.

Retention at Day 60: percentage of agents still active on day 60 (i.e., no cessation event before day 60), computed per run and then averaged across runs.

Time-to-dropout: day index of cessation; agents without cessation by day 60 are right-censored at day 60. Shaded bands in retention curves are plotted as mean ± 1.96 × SE, where SE=$\frac{s}{\sqrt{\{R\}}}$ from the run-level retention values at each day.

Hazard ratios are estimated using a Cox proportional hazards model fit to pooled traces, with time-to-dropout as the outcome and policy condition as the predictor within each cohort. Hazard ratio 95% CIs are computed from the model's standard errors on the log-hazard scale.

III. Results

We evaluated CARE-Sim under two feedback policies (**Rigid** vs **Compassionate**) using N = 400 unique agents (100 per cohort) and R = 5 independent 60-day runs per policy. Each run produces a complete 60-day record for every agent; we refer to each agent-by-run record as a trace. Accordingly, each policy yields 2,000 traces (400 agents × 5 runs), for a total of 4,000 traces across both policies. Because outcomes are generated by a seeded simulation (stressor schedules plus stochastic model sampling), we treat uncertainty summaries as measures of run-to-run stability rather than as population-level inferences. Unless noted otherwise, reported summary statistics are computed per run and then aggregated across the R = 5 runs. **Tbl. 1** summarizes Day-60 retention by cohort and coaching style, along with the between-condition advantage and the day on which retention curves first diverge by more than 5 percentage points. **Fig. 2** shows the corresponding retention trajectories over the full 60-day horizon.

A. Retention and the Emergent Shame Spiral

Fig. 2 shows retention over time by cohort and policy. Across runs, the **Resilient** cohort showed a ceiling effect, with >90% Day-60 retention under both policies, suggesting limited sensitivity to feedback tone among emotionally stable agents. By contrast, the largest separation emerged in the **Perfectionist** cohort. Under the **Rigid** policy, retention declined in a lagged pattern consistent with accumulating negative appraisal: repeated warning-style messages increased negative self-talk over time, reducing the internal motivation state and ultimately triggering dropout.

To quantify the separation, we fit a Cox proportional hazards model for **time-to-dropout** (day index of cessation), with right-censoring at day 60 for agents who remained active through the horizon. Within the Perfectionist cohort, the **Rigid** condition had a **dropout hazard ratio of 3.8** relative to the Compassionate condition (95% CI: 3.1–4.6), indicating a substantially higher risk of cessation under rigid feedback.

B. Sentiment Decoupling as a Mechanism

We next tested whether compassionate reframing changes the coupling between "doing poorly" and "feeling like a failure." We operationalized **dietary error** as the magnitude of daily caloric deviation from target (absolute deviation) and **negative self-talk** as the negative valence of the agent's reflective monologue scored via the VADER compound sentiment score.

Under the **Rigid** policy, dietary error and negative self-talk were strongly associated (**r = 0.82**), consistent with a tendency to globalize a lapse into moralized self-judgment. Under the **Compassionate** policy, the same association weakened substantially (**r = 0.35**), indicating that indicating that supportive reframing—indicating that supportive reframing—drawing loosely on principles of Cognitive Behavioral Therapy (CBT) without claiming to deliver clinical-grade therapy—reduced the tendency to equate a single-day error with self-worth. Mechanistically, this decoupling supports the survival-curve separation: compassionate prompts alter the affective update channel feeding the motivation state, rather than merely adding generic encouragement.

C. "What-the-Hell" Pattern Consistent with Reactance

The **Habitual/Defensive** cohort showed the clearest behavioral signature consistent with counter-regulatory responding. The Habitual/Defensive cohort experienced a dramatic early dropout rate followed by stabilization around Day 13. This occurs because agents in this cohort possess a high negative appraisal gain paired with lower initial motivation. Agents who encountered early stressors quickly depleted their motivation and dropped out. The surviving agents either faced favorable early stressor schedules or reached a stable behavioral equilibrium, keeping retention steady thereafter. We define a **Rigid "red warning" day** as any day t on which caloric intake exceeded target by **≥ 300 kcal** and the Rigid policy delivered its warning-form message (i.e., the message class logged as a rigid warning). We then computed the **next-day caloric change** as:

$$\%\Delta_{t\rightarrow t+1} = \frac{Calories_{\{t+1\}} - Calories_{\{t\}}}{Calories_{\{t\}}} \tag{5}$$

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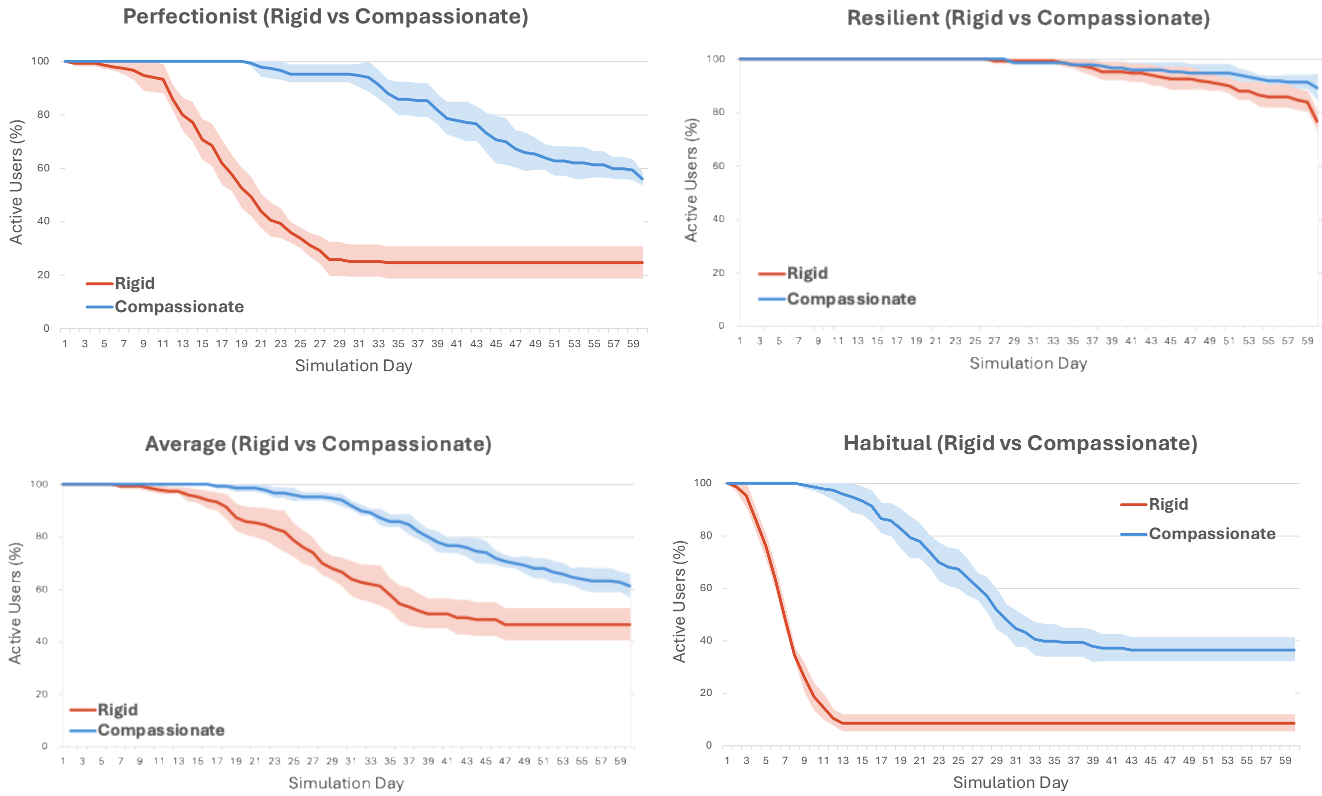

Fig. 2. User retention over 60 simulated days across four personality cohorts (Perfectionist, Resilient, Control, Habitual) under Rigid versus Compassionate coaching conditions. Solid lines represent the mean percentage of active agents (out of an initial N = 100 agents per cohort per run), averaged across R = 5 independent runs. Shaded bands indicate 95% uncertainty bands across runs (mean ± 1.96×SE). The Compassionate condition (blue) consistently outperformed the Rigid condition (red) in retaining users across all four personality types, though the magnitude and timing of divergence varied by cohort.

Relative to the cohort's average non-warning days, Habitual/Defensive agents exhibited a 12.4% increase in next-day intake following a red warning day. Notably, this pattern is not hard-coded (there is no rule of the form "if warned, eat more"). Instead, in the Habitual/Defensive configuration, rigid feedback functions as an additional stressor that depletes the internal willpower state used in the caloric decision, producing a trace pattern consistent with psychological reactance.

D. Consistency Checks and Qualitative Alignment

We conducted two sanity checks on the simulator outputs. First, run-to-run consistency in the relative ordering of cohort-by-policy outcomes was high (Run-to-run consistency in the relative ordering of cohort-by-policy outcomes was high (Intraclass Correlation Coefficient [ICC] = 0.88), computed from Day-60 retention values across repeated runs, suggesting that the observed separation patterns are robust across different stressor seeds.), suggesting that the observed separation patterns are not isolated artifacts of a single seed. Second, the overall retention trajectory exhibited a qualitative shape commonly observed in mHealth engagement (an early drop followed by a long tail). We treat this as **qualitative alignment**, not external validation; human-facing calibration remains future work.

IV. Discussion

Our findings suggest that feedback tone is not just a surface-level presentation choice; it can act as an interaction-policy stress multiplier whose effects depend strongly on user vulnerability. Compassionate coaching improved Day-60 retention relative to rigid coaching, with the largest advantages for the Perfectionist and Habitual/Defensive cohorts, while the Resilient cohort remained comparatively stable and diverged only late. This pattern suggests that warning-style feedback may be especially harmful for users who interpret setbacks through a self-critical lens. The retention differences support a human-factors interpretation in which a "failure" label can be informationally correct yet behaviorally counterproductive when it clashes with the user's appraisal style. In CARE-Sim, rigid feedback accelerated disengagement in vulnerability-typed agents, especially the Perfectionist cohort, where dropout risk was substantially higher. Mechanistically, this may reflect the system optimizing for compliance detection while the user experiences perceived judgment, turning a local lapse into a global self-assessment.

A key difference between coaching styles is not whether lapses occur, but how they propagate into affective state. Under

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rigid coaching, dietary error was tightly coupled with negative self-talk, whereas compassionate coaching weakened that coupling. Interpreted mechanistically, compassionate prompts act as a cognitive reframe: they preserve the informational content of a lapse while reducing identity-level judgment. When this decoupling holds, lapses remain recoverable, consistent with higher retention and a slower decline in motivation.

The Habitual cohort also showed a pattern consistent with counter-regulatory responding. Following rigid "red warning" days, next-day intake increased relative to baseline, and this was not hard-coded as a direct rule. Instead, it emerged from how rigid feedback altered downstream state updates and decision-making. This raises a practical concern: corrective messaging can sometimes become an additional trigger rather than a guide back to routine. Taken together, these results suggest a conservative design principle for diet-coaching systems: tone should be treated as a high-impact interaction parameter rather than a mere personalization preference. A rigid one-size-fits-all policy may be easy to implement, but it creates asymmetric harm risk for users high in perfectionism, self-criticism, or negative affectivity. In CARE-Sim, compassionate coaching appears to be the safer default, improving retention without changing tracking accuracy. In practice, systems could begin with nonjudgmental reframing and reserve more directive language for users who respond positively, using explicit opt-in or careful engagement monitoring.

A primary limitation of CARE-Sim is the circularity of validating LLM-generated policies against LLM-simulated users. Because both the coaching interventions and simulated users are driven by GPT-4o, the observed dynamics partly reflect model alignment with its own prompt instructions rather than independent human behavior. This is consistent with critiques of silicon sampling and generative agents, which caution against treating synthetic text generation as a direct proxy for human psychological processes.

Accordingly, CARE-Sim should not be read as evidence that real users will reproduce these exact retention curves. Instead, it functions as an upper-bound test of interaction policy efficacy: if a policy fails to produce a theoretically coherent trajectory in a constrained, compliant simulation environment, it is unlikely to succeed in the noisier reality of deployment. The numerical outputs, including hazard ratios, should therefore be interpreted as measures of comparative risk under controlled simulation assumptions, not as transfer-ready human effects.

The next step is to strengthen external anchors and isolate mechanisms. High-leverage follow-on work includes ablation studies that hold message length and actionability constant while varying tone, plus lightweight human-facing validation such as vignette-based ratings of perceived judgment, support, and disengagement intention. A complementary direction is calibration against coarse real-world engagement statistics so that simulated trajectories better reflect realistic attrition regimes before policy comparisons are interpreted.

These ethical considerations remain important even for simulation work. CARE-Sim is intended to reduce exposure to potentially harmful punitive conditions during early-stage policy evaluation, but its findings should not be used to justify coercive or manipulative messaging in deployed systems. For practitioners, CARE-Sim offers a way to pre-screen prompt policies against synthetic vulnerable populations before A/B testing on real users. If the simulator predicts high attrition for a policy, developers can refine the tone in silico without risking harm to human subjects. If a policy performs well in simulation, the next step should be user-centered validation, such as vignette-based studies in which participants rate the perceived judgment and supportiveness of the responses. That approach lets you test whether simulated empathy translates to human perception without exposing users to a live, potentially punitive intervention.

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